A Variable Definitions

This appendix lists every variable used in the empirical analyses, with construction formula, data source, and originating reference. Variable names follow paper-standard conventions where available (Dzieliński et al. 2021 for speech measures, Bates et al. 2009 for cash-holdings controls); deviations from source-paper frequency or sample are documented in the formula column.

A.1 Sample Manifest

Name	Description / Formula	Source	Reference
`file_name`, `ceo_id`, `ceo_name`, `gvkey`, `ff12_code`, `ff12_name`, `start_date`	Master sample manifest with CEO, firm, and industry identifiers	outputs/1.4_AssembleManifest	

A.2 Text/Linguistic Variables

Name	Description / Formula	Source	Reference
`ClarityCEO`	CEO clarity persistent-style component (DWZ Eqn 5). *Formula:* $\text{ClarityCEO}_i = -\widehat{\text{FE}}_i$ where $\widehat{\text{FE}}_i$ is the CEO-i fixed-effect estimate from the panel regression $\text{UncAnsCEO}_{ic} = \alpha + \sum_i \text{FE}_i \cdot \mathbb{1}[\text{CEO} = i] + \mathbf{X}_{ic}\gamma + \varepsilon_{ic}$ over the call-level Q&A uncertainty panel; controls $\mathbf{X}$ are the DWZ extended-controls vector. The sign flip on $\widehat{\text{FE}}_i$ makes higher ClarityCEO mean "CEO speaks more clearly" (lower mean uncertainty). Persistent CEO trait.	.../ceo_clarity_extended/	Dzieliński et al. (2021) (Section 4.4 Eqn 5, p.16)
`ClarityCEO_QtrExp`	ClarityCEO under within-tenure expanding window. *Formula:* As ClarityCEO, but the CEO fixed-effect regression is re-estimated on an expanding within-CEO window using only calls of CEO i available up to and including focal call c. Strictly look-ahead-free.	.../ceo_clarity_expanding/	thesis QtrExp variant of Dzieliński et al. (2021) Eqn 5
`NegCall`	Entire-call negative sentiment percentage. *Formula:* Ratio of negative words to total words in the entire call, based on LM 2011 negative-words list. Per DWZ 2021 Appendix Table A.1: 'percentage of negative words in all words spoken by the CEO, CFO and analysts attending the call.'	outputs/2_Textual_Analysis/2.2_Variables	Dzieliński et al. (2021) (Section 4.4, p.18; Appendix Table A.1, p.54)
`UncAnsCEO`	CEO-only Q&A uncertainty percentage. *Formula:* CEO Q&A uncertainty (%) = UnctWordsAnsCEO / WordsAnsCEO (DWZ 2021 Eqn 2). Uncertainty wordlist = LM 2011 Master Dictionary 'uncertainty' list (297 words, Aug 2014 version). CEO identified via Execucomp ceoann field.	outputs/2_Textual_Analysis/2.2_Variables	Dzieliński et al. (2021) (Section 4.2, Eqn 2, p.13; Appendix Table A.1, p.54)
`UncPreCEO`	CEO-only presentation uncertainty percentage. *Formula:* CEO Presentation uncertainty (%) = UnctWordsPreCEO / WordsPreCEO (DWZ 2021 Eqn 1). Uncertainty wordlist = LM 2011 (297 words).	outputs/2_Textual_Analysis/2.2_Variables	Dzieliński et al. (2021) (Section 4.2, Eqn 1, p.13; Appendix Table A.1, p.54)
`UncQue`	Analyst Q&A uncertainty percentage. *Formula:* Analyst Q&A uncertainty (%) = UnctWordsQue / WordsQue (DWZ 2021 Eqn 3). Uncertainty wordlist = LM 2011 (297 words).	outputs/2_Textual_Analysis/2.2_Variables	Dzieliński et al. (2021) (Section 4.2, Eqn 3, p.13; Appendix Table A.1, p.54)
`UncResCEO`	CEO Q\&A residual call-state uncertainty component (DWZ Eqn 4). *Formula:* $\widehat{\varepsilon}_{ic}$ = the residual from the panel regression $\text{UncAnsCEO}_{ic} = \alpha + \sum_i \text{FE}_i \cdot \mathbb{1}[\text{CEO} = i] + \mathbf{X}_{ic}\gamma + \varepsilon_{ic}$ that produces ClarityCEO. By OLS first-order conditions, the within-CEO mean of UncResCEO is exactly zero by construction. Captures call-level deviation in CEO Q&A uncertainty after stripping out the persistent CEO mean.	.../ceo_clarity_extended/	Dzieliński et al. (2021) (Section 4.4 Eqn 4, p.16)
`UncResCEO_QtrExp`	UncResCEO under within-tenure expanding window. *Formula:* As UncResCEO, but the residualization regression is re-estimated on an expanding within-CEO window. Strictly look-ahead-free.	.../ceo_clarity_expanding/	thesis QtrExp variant of Dzieliński et al. (2021) Eqn 4

A.3 Financial / Market Variables

Name	Description / Formula	Source	Reference
BGTLevel_Spread	Compute BGT (2018) long-window closing bid-ask spread around earnings calls. *Formula:* CRSP via get_raw_daily_data (closing BID/ASK) (see module: src/f1d/shared/variables/bgt_long_window_spread.py)	CRSPviaget_raw_daily_data	Bushee, Gow & Taylor (2018, JAR) [window]; Lee (2016, TAR) [formula]
Capex	Build Capex = capxy_annual / atq_lag from raw Compustat quarterly data. *Formula:* Compustat/capxy_Q4/atq (see module: src/f1d/shared/variables/capex_intensity.py)	Compustat/capxy_Q4/atq	Bates, Kahle & Stulz (2009, JF)
CashFlowAt	Build CashFlow = oancfy / atq from raw Compustat quarterly data. *Formula:* Compustat/oancfy/atq (see module: src/f1d/shared/variables/cash_flow.py)	Compustat/oancfy/atq	Bates, Kahle & Stulz (2009, JF)
CashRatio	Build CashRatio = cheq / atq from raw Compustat quarterly data. *Formula:* Compustat/cheq/atq (see module: src/f1d/shared/variables/cash_holdings.py)	Compustat/cheq/atq	Bates et al. (2009) (Section II, p.1991; BKS use annual che/at — we use quarterly cheq/atq for panel alignment)
DailyVola	Annualized stock-return volatility computed on the inter-call CRSP daily-return window. *Formula:* Annualized standard deviation of CRSP daily returns over the inter-call window [prev_call_date + 5 days, call start_date - 5 days], requiring >= 10 trading days. Formula: std(daily_ret) * sqrt(252) * 100, expressed in percent.	CRSP/RET	Dzielínski et al. (2021)
DivDummy	Build DivDummy = (dvy > 0).astype(float) from raw Compustat quarterly data. *Formula:* Compustat/dvy>0 (see module: src/f1d/shared/variables/dividend_payer.py)	Compustat/dvy>0	Bates et al. (2009) (Section IV, p.2000; dividend payout dummy based on common dividends)
EarnVol	Builder for Earnings Volatility (earnings_volatility) variable. *Formula:* Compustat (see module: src/f1d/shared/variables/earnings_volatility.py)	Compustat	Duong et al. (2025) (Appendix B, p.30; 3-year rolling std of qoq asset-scaled earnings)
FirmMat	Builder for Firm Maturity (firm_maturity) variable. *Formula:* Compustat (see module: src/f1d/shared/variables/firm_maturity.py)	Compustat	Biddle et al. (2009) (Appendix A, p.130; years since first CRSP appearance); alt Leary and Roberts (2010) (App.~A p.354)
Leverage	Build Leverage = (dlcq + dlttq) / atq (interest-bearing debt only). *Formula:* Compustat/dlcq,dlttq,atq (see module: src/f1d/shared/variables/book_lev.py)	Compustat/dlcq,dlttq,atq	Bates et al. (2009), rz1995
lnAssets	Build Size = ln(atq) from raw Compustat quarterly data. *Formula:* Compustat/atq (see module: src/f1d/shared/variables/size.py)	Compustat/atq	Dzielínski et al. (2021)
PRisk	Match quarterly PRisk onto each earnings call by (gvkey, calendar_quarter). *Formula:* Hassan et al. (2019) quarterly PRisk (see module: src/f1d/shared/variables/prisk_q.py)	Hassanetal.	Hassan et al. (2020)
RDSales	Build RDSales = xrdq / atq from raw Compustat quarterly data. *Formula:* Compustat/xrdq,atq (see module: src/f1d/shared/variables/rd_intensity.py)	Compustat/xrdq,atq	Bates et al. (2009), jjl2021
ROA	Build ROA = iby_annual / avg_assets from Compustat quarterly data. *Formula:* Compustat/iby,atq (see module: src/f1d/shared/variables/roa.py)	Compustat/iby,atq	Wang (2020), Dzielínski et al. (2021)
SalesGrowth	Build SalesGrowth from raw Compustat quarterly data (annual Q4 panel). *Formula:* Compustat/saley_Q4_YoY (see module: src/f1d/shared/variables/sales_growth.py)	Compustat/saley_Q4_YoY	Biddle et al. (2009) (Section 3.2, p.117; pct change in sales)

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Name	Description / Formula	Source	Reference
sCFO	Build sCFO = rolling 5-year std (min 3 yrs) of (oancfy/atq_{t-1}) per gvkey. *Formula:* Compustat/oancfy/atq_{t-1} rolling std (see module: src/f1d/shared/variables/ocf_volatility.py)	Compustat/oancfy/atq_{t-1}rollingstd	Biddle et al. (2009) (Appendix A, p.130; 5-year rolling std of CFO/avg assets)
StockPrice	Compute stock price at earnings call date (VECTORIZED). *Formula:* CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/stock_price.py)	CRSPviaget_raw_daily_data	Amiram et al. (2016) (Appendix A, p.137; daily CRSP stock price; justification cites Stoll 1978)
SurpDec	Signed-quintile earnings surprise rank (-5 to +5) within calendar quarter. *Formula:* Signed quintile rank of percentage earnings surprise per DWZ 2021 Appendix Table A.1: raw surprise = (actual - consensus forecast earnings) / share price 5 trading days before announcement, x 100. Firms grouped into 5 bins of positive surprise (+5 largest through +1 smallest positive), 0 for zero surprises, and 5 bins of negative surprise (-1 smallest negative through -5 largest negative). DWZ footnote 32 notes this decile convention follows Hirshleifer, Lim & Teoh (2009) and DellaVigna & Pollet (2009).	.../variables/earnings_surprise.py	Dzieliński et al. (2021) (Appendix Table A.1, p.55; convention cites DellaVigna-Pollet 2009 + Hirshleifer-Lim-Teoh 2009)
TobinsQ	Build TobinsQ = (AT + cshoq*prccq - CEQ) / AT from raw Compustat quarterly data. *Formula:* Compustat/(atq+cshoq*prccq-ceqq)/atq (see module: src/f1d/shared/variables/tobins_q.py)	Compustat/	Bates, Kahle & Stulz (2009, JF)
Turnover	Compute share turnover at earnings call date (VECTORIZED). *Formula:* CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/turnover.py)	CRSPviaget_raw_daily_data	Amihud (2002) (Section 2.1, p.33; alt rolling-window def in Chang et al. (2006) Appendix B p.3045)

A.4 Econometric / Indicator Variables

Name	Description / Formula	Source	Reference
HighCFvol	High cash-flow-volatility moderator (H1.3). *Formula:* Binary = 1 if firm-quarter Han-Qiu (2007) cash-flow-volatility (16-quarter rolling std of OCF / |mean| of OCF) is at or above the within-Fama-French-12-industry-year median; 0 otherwise.	runtime	Han and Qiu (2007); thesis (H1.3 split convention)
Lagged_DV	Generic lagged-DV control (1Q lag of suite-specific DV). *Formula:* 1-quarter lag of the dependent variable, constructed at runtime per spec (DV varies per suite). Always included as base control to absorb persistence.	runtime	thesis convention (per feedback_lagged_dv.md)
Unrated	No-rating subsample indicator (H1.2). *Formula:* Binary = 1 if firm has no S&P rating at call date; 0 otherwise.	runtime	thesis (H1.2 rating subsample)
z_log_TotalSimilarity	z-scored log of HP 2016 product-market-similarity (IV in H23 competition test). *Formula:* z-score of log(1 + TotalSimilarity) computed within year. TotalSimilarity = HP 2016 firm-year product market similarity score (sum of pairwise cosine similarities × 100).	runtime	Hoberg and Phillips (2016)

A.5 Runtime Derivatives (Lead/Lag/Centered/Interaction)

Name	Description / Formula	Source	Reference
AbsSurpDec	Absolute earnings surprise quintile magnitude. *Formula:* abs(SurpDec) — unsigned magnitude of signed-quintile earnings surprise rank, values 0-5.	runtime	thesis (magnitude control independent of sign)

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Name	Description / Formula	Source	Reference
BGTLevel_Spread_lead1	1Q lead of BGTLevel_Spread. *Formula:* 1-quarter lead of BGTLevel_Spread	runtime	Bushee et al. (2018), lee2016
CashRatio_lead	1Q lead of CashRatio. *Formula:* 1-quarter lead of CashRatio	runtime	see base var
ClarityCEO_c	Mean-centered ClarityCEO. *Formula:* ClarityCEO minus its sample mean (mean-centered for interaction interpretability).	runtime	thesis (interaction interpretability)
ClarityCEO_QtrExp_c	Mean-centered ClarityCEO_QtrExp. *Formula:* ClarityCEO_QtrExp minus its sample mean.	runtime	thesis (interaction interpretability)
PRisk_lag2	2Q lag of PRisk. *Formula:* 2-quarter lag of PRisk	runtime	Hassan et al. (2020)
UncPreCEO_c	Mean-centered UncPreCEO. *Formula:* UncPreCEO minus its sample mean (mean-centered for interaction interpretability).	runtime	thesis (interaction interpretability)
UncPreCEO_c_x_HighCFvol	Interaction: UncPreCEO_c $\times$ HighCFvol (H1.3). *Formula:* Product: UncPreCEO_c * HighCFvol.	runtime	thesis (CFvol moderator interaction)
UncPreCEO_c_x_Unrated	Interaction: UncPreCEO_c $\times$ Unrated (HFC). *Formula:* Product: UncPreCEO_c * Unrated.	runtime	thesis (HFC interaction)
UncResCEO_c	Mean-centered UncResCEO. *Formula:* UncResCEO minus its sample mean (mean-centered for interaction interpretability).	runtime	thesis (interaction interpretability)
UncResCEO_c_x_HighCFvol	Interaction: UncResCEO_c $\times$ HighCFvol (H1.3). *Formula:* Product: UncResCEO_c * HighCFvol.	runtime	thesis (CFvol moderator interaction)
UncResCEO_c_x_Unrated	Interaction: UncResCEO_c $\times$ Unrated (HFC). *Formula:* Product: UncResCEO_c * Unrated.	runtime	thesis (HFC interaction)
UncResCEO_QtrExp_c	Mean-centered UncResCEO_QtrExp. *Formula:* UncResCEO_QtrExp minus its sample mean.	runtime	thesis (interaction interpretability)
UncResCEO_QtrExp_c_x_Unrated	Interaction: UncResCEO_QtrExp_c $\times$ Unrated (HFC QtrExp). *Formula:* Product: UncResCEO_QtrExp_c * Unrated.	runtime	thesis (HFC QtrExp interaction)

A.6 Macro / Regulatory Variables

Name	Description / Formula	Source	Reference
CCCL	SEC comment letter received (binary, forward-looking) *Formula:* 1 if SEC comment letter received in next quarter, 0 otherwise	inputs/EDGAR_CommentLetters	cdm2013
GEPU_log	Davis (2016, NBER WP 22740) Global Economic Policy Uncertainty (GEPU) index (log-transformed, current-$ GDP weights) *Formula:* log(GEPU_current) — GDP-weighted average of 16 country EPU indices, matched by calendar month of call start_date	inputs/EconomicPolicyUncertaintyIndex/Global	Davis (2016)

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Name	Description / Formula	Source	Reference
`US_EPU_log`	Baker, Bloom & Davis (2016, QJE) US News-Based Economic Policy Uncertainty index (log-transformed) *Formula:* log(News_Based_Policy_Uncert_Index) matched by calendar month of call start_date	inputs/EconomicPolicyUncertaintyIndex/US	Baker et al. (2016)

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